

measures to ensure that their electronic equipment function properly even under demanding conditions. This is especially true for assemblies which are incorporated in aerospace and military, medical and also automotive systems (engine control, electronic central locking, electric windows and air-bags). In many cases, worldwide marketing of electronic components is possible only if these have a protective coating.

Coatings are employed for different requirements of the system:

1. Mechanical protection, e.g., against mechanical abrasion, vibration, minor impact loads
2. Corrosion protection, e.g., against atmospheric humidity, dew-point condensation, hand perspiration and diverse chemicals
3. Protection against mould infestation when employed under tropical conditions, for minimising dendritic growth
4. Improvement of electrical insulation between two potentials, e.g., in case of requirements for explosion protection and in case of high-voltage stress, particularly at greater heights

In case of higher stress, particularly aggressive industrial atmospheres and exposure to material such as fuels, oils and lubricants, a one-component conformal coating no longer offers sufficient resistance; two-pack conformal coatings or casting compounds are employed to protect such electronics.

In practice, electronic assemblies are always more or less tolerant to normal range of humidity. Under 'normal' circumstances, i.e., approx. 40 per cent to 70 per cent R.H., a problem-free operation is normally ensured. At a humidity level above 70 per cent, proper operation becomes critical. Increasingly, humidity absorp-

tion becomes negatively noticeable in the form of possible leakage currents. The first signs of dewing behaviour may appear because the critical relative humidity of many salts has already been reached. For a reliable operation with the possible danger of dewing, a protective coating is advisable. (See Fig. 1)

The possibility or necessity for a protective coating should be considered during the planning phase, or it should be incorporated at an early stage into the design of the assembly. Care has to be taken here in the choice of components with respect to their coatability and layout specifications for the coating process. The latter can include space requirements for non-coatable areas as well as arrangement of edge connectors.

In this connection, views concerning the occurrence and remedy of errors during a product life-cycle are important. It is said that 75 per cent of errors occur during the planning

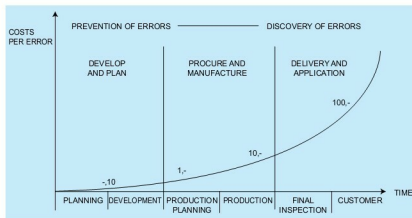


Fig. 3: Decadic law of error costs

and development phase, whereas 80 per cent of the errors are remedied only during the final inspection or during operation. (See Fig. 2)

Of further consideration is the so-called 'decadic law of error costs,' which states that the error costs increase by a factor of 'ten' in every phase of the product life-cycle. (See Fig. 3)

The cost-driving variables can be, for example, non-optimised coating processes such as layout-related complicated selective coating instead of cost-effective dip coating in the multiple-use carrier. In particular, possible incompatibilities and their elimination from up- or downstream processes of the coating process can also assist in the run-up, in order to keep coating costs down.

In this context, it should also be observed that only a well-applied coating that has a verified compatibility to preliminary processes provides high-quality protection to the electronic assembly against the loads expected.

Requirements set upon coated assemblies

Requirements concerning climate resistance coupled to mechanical and/or electrical properties have to be maintained, in relation to their protective effect, under the most diverse climates for electronic assemblies.

TABLE I
Properties of Coating Materials

Electrical properties	Thermal properties	Constructive properties	Chemical and physical properties
➤ Resistivity ➤ Surface resistivity ➤ Tracking resistance ➤ Dielectric strength ➤ Dielectric loss factor ➤ Dielectric constant ➤ Moisture and insulation resistance	➤ Coefficient of thermal expansion ➤ Thermal conductivity ➤ Glass transition temperature ➤ Temperature resistance ➤ Inflammability ➤ Dimensional stability ➤ Optical properties (discolouration)	➤ Processability ➤ Adhesion ➤ Tensile strength ➤ Compressive strength ➤ Flexural strength ➤ Modulus of elasticity ➤ Impact strength ➤ Hardness	➤ Water absorption ➤ Permeability ➤ Chemical resistance ➤ Tropic resistance ➤ Hydrolytic resistance ➤ Climatic resistance ➤ Light-/UV-resistance